

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 01



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Topics to be Covered



Topic-1

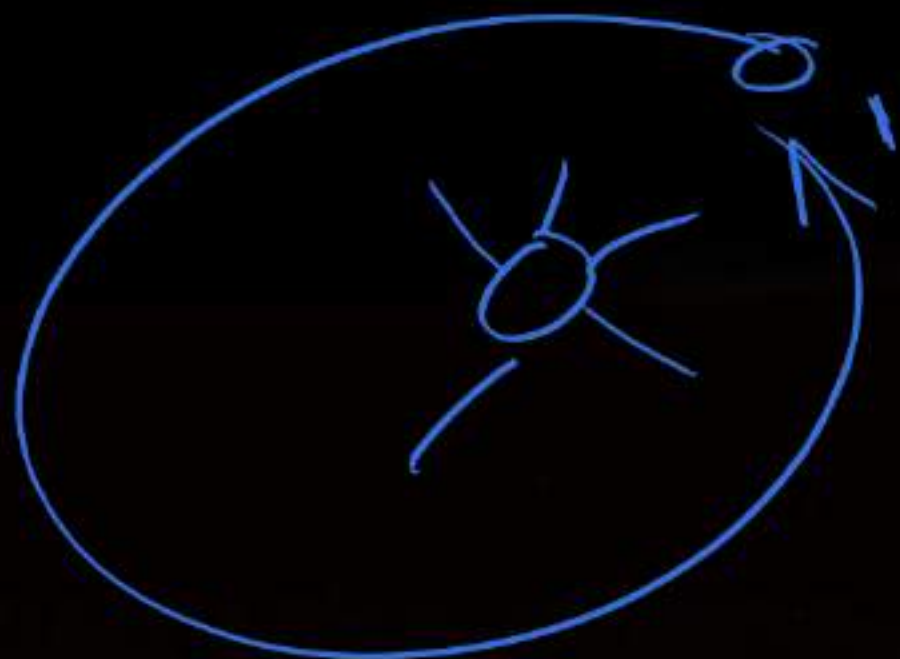
Calendar ✓

ODD DAYS



CALENDAR

SOLAR ✓



Basic Questions



1. Use of CALENDAR?

MONDAY
Date → Day?

01-01-01

Common Era

2. How many months in a year consist of 30 days?

only

11 months

3. Which is the first day of a week?

SUNDAY

4 months

4. What is the difference between A.D. & B.C.?

00-00-00

Before Christ

(Holy Year) Anno Domini

1542 C.E.

B.C.E.

April

June

Sep

Nov

Before
Common
Era

LEAP YEAR Occurrence



400
Maximum $\rightarrow$ 8 years

16 41 $\times$

16 32 $\checkmark$

12 18 $\times$

29th Feb

1996 $\rightarrow$ 29th Feb

2000 $\rightarrow$ 29th Feb

8 yrs

1796 $\rightarrow$ 29th Feb

? 1800 $\times$

1804 $\rightarrow$ 29th Feb

To be Noted:

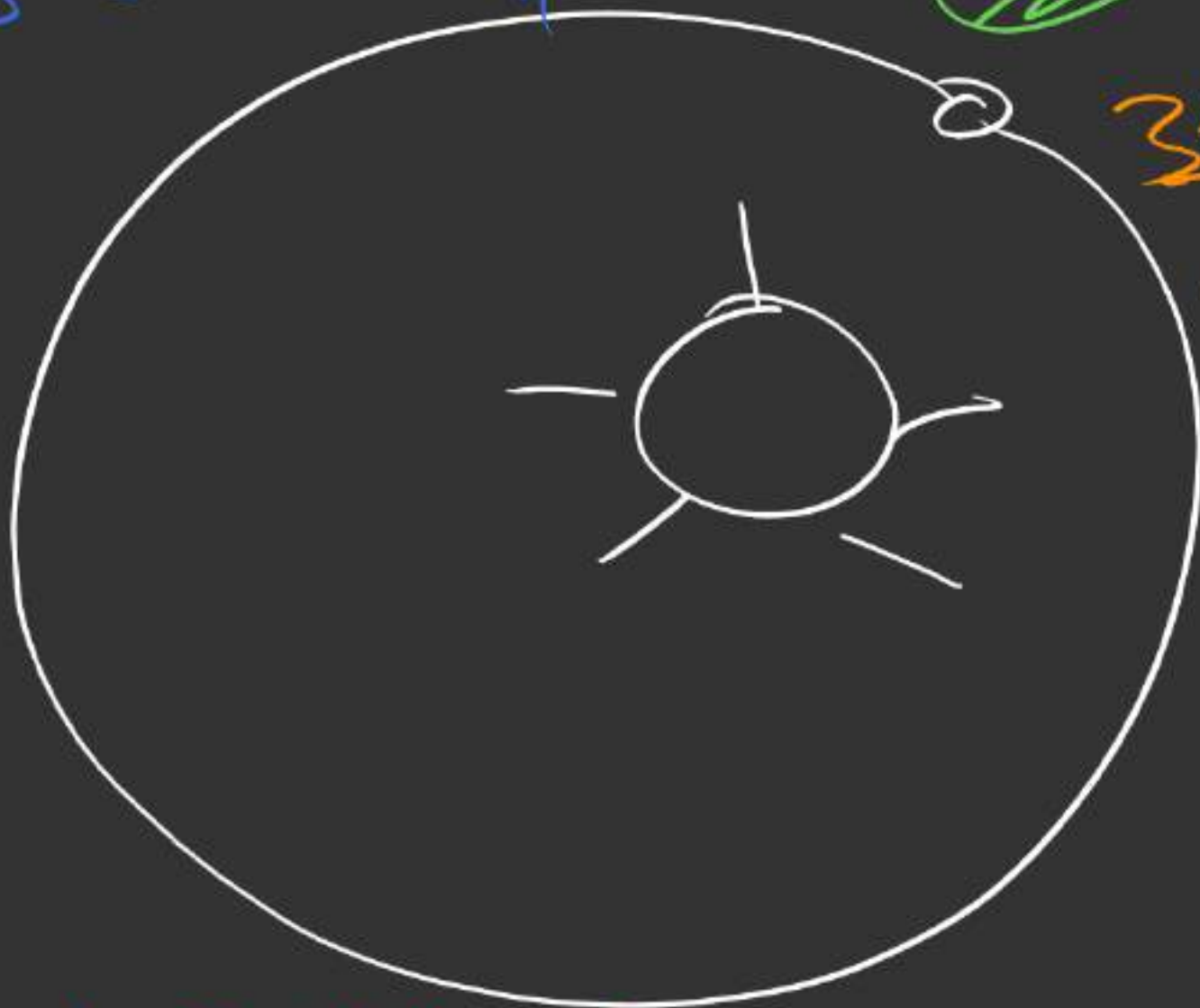
- ✓ If a year is divisible by 4 then it's a leap year else normal. e.g. 1652, 1212, 1496, 1708, etc. are leap years whereas 1714, 1446, 2006 etc. are normal years.
- ✓ If a century is divisible by 400 then it's a leap year else normal. e.g. 1600, 1200, 2000, 800 etc. are leap years whereas 1000, 1500, 2100 etc. are normal years.

365 days shows 42 min --- 500ⁿ

500ⁿ

365 $\frac{1}{4}$ days

365 days 6 hours



1 $\rightarrow$ 365

1 $\rightarrow$ 365

1 $\rightarrow$ 365

1 $\rightarrow$ 365 + 1 = 366 days L.Y.

24 hrs
or
1 day

6 hrs
6 hrs
6 hrs
6 hrs

ODD DAYS

As today is 21st November, 2025. The day is Friday.

✓ What was the day on 11th November, 2025?

-3

→ Tuesday

7) 60 days (8 weeks)

56

—

4 odd days

10 days

3 days

7

ODD DAYS



$$\begin{array}{r} 7 \overline{) 40} \\ \underline{35} \\ 5 \end{array}$$

Handwritten calculation showing 40 divided by 7, resulting in a remainder of 5. The word "Odd" is written in red and circled in blue.



✓ All those number of days which can't be kept in a group of a week.

OR

✓ When given number of days is divided by 7, the remainder is called as odd days.

Normal Year & Leap Year

~~52 weeks~~

100 years $\rightarrow$ 5 odd days

L.Y. 24
N.Y. 76

(x2) 48
(x1) 76

6 + 6 = 12

N.Y. $\rightarrow$ 365 $\rightarrow$ 1 odd day

L.Y. $\rightarrow$ 366 $\Rightarrow$ 2 odd days

x3
04✓
08✓
12✓
16
20
24
28
32
36
40
44
48
52
56
60
64
68
72
76
80
84
88
92
96
100

(x24)

First Century

500	$\times 100 \rightarrow$	<u>5</u> odd days	900	1300	1700
600 (10)	$\times 200 \rightarrow$	<u>3</u> odd days	1000	1400	1800
700 (15)	$\times 300 \rightarrow$	<u>1</u> odd day	1100	1500	1900
800 (20)	$\times 400 \rightarrow$	<u>0</u> odd days	1200	1600	2000

$\text{century} \% 4 == 0 \Rightarrow 0 \text{ odd days}$
 $(\text{century} - 100) \% 4 == 0 \Rightarrow 5 \text{ odd days}$
 $(\text{century} - 200) \% 4 == 0 \Rightarrow 3 \text{ odd days}$
 $(\text{century} - 300) \% 4 == 0 \Rightarrow 1 \text{ odd day}$

century == given year in xx00 format
 century % 4 == 0 => 0 odd days
 (century - 100) % 4 == 0 => 5 odd days
 (century - 200) % 4 == 0 => 3 odd days
 (century - 300) % 4 == 0 => 1 odd day

Questionnaire:



SATURDAY 6 odd days
27th July, 2002

0 → Sun
1 → Mon
2 → Tue
3 → Wed
4 → Thurs
5 → Fri
6 → Sat

2000 → 0
1 → 1⁺

2001 → 1 (1 → 3)
5 → 2

(31) J → ~~3~~ (27) J → ~~6~~

(28) F → 0

M → ~~3~~

A → ~~2~~ +

5 odd days

Questionnaire:



TUESDAY

2022 days

3rd June, 1947

46 → 1
├── L: 4
│ 11
│ 22
│ ①
└── N: 4
 35
 35
 ⑥

1900 → 1
+
46 → 1

1946 → 2 + 0

J - 3 M - 3
F - 0 J - 3
M - 3 0
A - 2 X

04
08
12
16
20
⋮
44
N

Questionnaire:

Thursday 4 odd days

#Q. Guru Nanak was born on 15th April, 1469. What was the week day?

$$\begin{array}{r} 1400 \rightarrow 3 \\ + \\ 68 \rightarrow 1 \\ \hline \end{array}$$

$$1468 \rightarrow 4$$

$$\begin{array}{l} J - 3 \\ F - 0 \\ M - 3 \\ A \rightarrow 1 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 68 \rightarrow 1 \\ \hline L.Y \quad N.Y \\ 17 \quad 51 \\ \hline 34 \quad 51 \end{array}$$

$$6 + 2 = 8$$

$$\begin{array}{r} 64 \\ \times 17 \\ \hline 68 \end{array}$$

Fact:



April 15, 1469: Day of the Week

April 15, 1469 was the 105th day of the year 1469 in the Gregorian calendar. There were 260 days remaining until the end of the year. The day of the week was Thursday.

The day of the week for April 15, 1469 under the old Julian calendar was Saturday. Did you notice the difference with the Gregorian calendar?

Questionnaire:

Thursday
26th January, 1950

$$\sqrt{1900} \Rightarrow 1$$

$$+ 49 \Rightarrow 5$$

C.Y N.Y

12 37
 24 37

$$3 + 2 = 5$$

- (A) SUNDAY
- (B) FRIDAY
- (C) WEDNESDAY
- (D) None

$$1949 \Rightarrow 6$$

$$Jan(26) \Rightarrow 5$$

$$\Rightarrow 4 \text{ odd days}$$

16 Feb, 2004 MONDAY

2000 → 0
3 → 3

2003 → 3⁰⁰

J - (3)

F → (2)

5e

8

↓

1000 DAY

[MCQ]

Assignment



#Q. Mahatma Gandhi was born on 2nd October 1869. What was the week Day?

- A** Monday
- B** Thursday
- C** Tuesday
- D** Saturday

[MCQ]

Assignment



#Q. If January 1st, 1992 was a Wednesday. What day of the week was January 1st, 2003?

- A** Sunday
- B** Wednesday
- C** Thursday
- D** Friday



2 mins Summary



Topic

Calendar

odd days

^u
Date → Day?

THANK - YOU